RESEARCH ETHICS & STANDARDS

Ethics principles & guidelines • Originality & plagiarism • Societal & sustainability aspects

Tanvirul Islam
Lecturer
Department of CSE, DIU

LEARNING OUTCOMES

- Apply ethics principles to common CS research situations
- Distinguish fabrication, falsification, plagiarism, and “questionable practices”
- Write without plagiarism (text, code, figures, data) using a safe workflow
- Identify societal risks (bias, privacy, misuse) and propose mitigations
- Report compute responsibly and design more sustainable experiments

WHAT "RESEARCH ETHICS" MEANS

In CS, ethics commonly appears in:

- **Data** (privacy, consent, bias, licensing)
- **Experiments** (fair evaluation, no leakage, honest reporting)
- **Writing** (proper credit, no plagiarism)
- **Release** (misuse risks, security, fairness)
- **Compute** (cost, energy, sustainability)

THE 5-PRINCIPLE ETHICS CHECKLIST

Use this before collecting data, running experiments, or publishing:

- **Honesty** — report results truthfully (including failures)
- **Respect** — protect privacy, consent, dignity
- **Fairness/Justice** — avoid discrimination; don't shift harms to vulnerable groups
- **Responsibility** — anticipate misuse and safety risks
- **Stewardship** — handle data/resources securely and sustainably

1-question test:
"If this decision was public, could I defend it clearly?"

RESEARCH MISCONDUCT (WHAT NOT TO DO)

Fabrication: making up data/results that never happened

- Example: "We ran 50 trials" but ran only 10

Falsification: changing or hiding parts of the process/data to force a result

- Example: removing "hard" test cases without reporting

Plagiarism: presenting others' work as yours

- Includes text, code, figures, datasets, and ideas

Also common in CS (Questionable Practices):

- Test-set tuning ("benchmark gaming")
- Reporting only the best run and hiding failures
- Changing the metric after seeing results

ETHICS “GUIDELINES” VS “STANDARDS”

- **Guidelines** = recommended best practices (how to do it well)
- **Standards** = expected rules for credibility (what good research must satisfy)

In CS, standards usually require:

- Clear **problem definition** + assumptions
- Fair **baselines** and metrics
- **Train/validation/test** separation
- Reproducibility (enough detail to repeat)
- Proper **citations** (datasets/tools/ideas)
- Ethical data handling (privacy, licenses)

REPRODUCIBILITY CHECKLIST (WHAT TO INCLUDE IN YOUR PAPER)

To make your work reproducible, include:

- Dataset details (version, preprocessing, filtering rules)
- Model/algorithm description + hyperparameters
- Evaluation protocol (splits, metrics, seeds, number of runs)
- Baselines (which ones, how configured)
- Compute details (hardware type, runtime, training steps)
- Limitations and failure cases

HUMAN SUBJECTS & PRIVACY (WHEN ETHICS APPROVAL MATTERS)

You have human-subject / personal data if it relates to people, e.g.:

- Surveys, interviews, user studies
- Logs (clicks, chats), location traces
- Faces, voices, biometrics
- Student records, health-related texts
- “Public” social media data (still about people)

Minimum ethical actions:

- Data minimization (collect only what you need)
- Secure storage (access control; encryption if possible)
- De-identification (but remember re-identification risk)
- Consent when feasible; extra care for vulnerable groups

CS warning: “Anonymous” logs can still be re-identified by combining signals.

DATA ETHICS: CAN I USE THIS DATASET?

Before using a dataset, answer these 5:

1. **Source:** Where did it come from?
2. **Permission:** What does the license/terms allow?
3. **Privacy:** Any personal/sensitive data?
4. **Bias:** Who is underrepresented or missing?
5. **Security:** Could it expose vulnerabilities or identities?

Example:

Scraping a site for training data may violate terms, contain personal info, or include copyrighted content.

AUTHORSHIP & CREDIT (STANDARD PRACTICE)

Authorship should match contribution, not seniority.

Common contribution types:

- Idea/design, implementation, experiments, analysis, writing, supervision

Unethical patterns:

- **Gift authorship:** adding someone who didn’t contribute
- **Ghost authorship:** excluding a real contributor
- **Credit theft:** taking student/junior work without recognition

PEER REVIEW & CONFLICTS OF INTEREST (COI)

Conflict of interest = your judgment could be biased

Examples:

- *reviewing a friend's paper*
- *reviewing a competitor's paper*
- *financial/company incentive*

Ethical review rules:

Confidentiality (don't share the paper)

Don't use ideas/code before it's public

Give constructive, specific feedback

Analogy:

Peer review is like a security audit—never exploit what you learn.

WHAT “ORIGINALITY” MEANS IN CS

Originality can be:

- **New method** (algorithm/model/system)
- **New setting** (edge devices, privacy constraints, real-time)
- **New dataset/benchmark**
- **New insight** (failure mode, analysis, theory)
- **New engineering** (reproducible pipeline, system optimization)

Key rule:
Using existing tools is normal; ***claiming them as your invention is not.***

PLAGIARISM

Plagiarism includes:

- **Text plagiarism** (copying sentences/paragraphs)
- **Mosaic plagiarism** (patchwork paraphrasing)
- **Idea plagiarism** (using a core idea without credit)
- **Figure plagiarism** (reusing diagrams/images without attribution/permission)
- **Code plagiarism** (copying code without citation + license compliance)
- **Self-plagiarism** (reusing your old writing/results as “new” without disclosure)

***Important: code + data + figures** are academic content too.*

“IS THIS OK?” QUICK EXAMPLES

Decide: **OK** / **Not OK** / **Depends**

1. Reuse your previous paper’s paragraph verbatim, no disclosure
2. Paraphrase baseline description + cite
3. Use a figure from a paper with citation
4. Use AI to fix grammar only

HOW TO AVOID PLAGIARISM

Practical workflow:

1. Read sources → write notes in your own words
2. Draft from notes (not by editing the original text)
3. Add citations while writing (don't postpone)
4. Track sources per section (Methods, Dataset, Baselines…)
5. Final check: "Can a reader trace every key claim?"

Always cite: datasets, benchmarks, libraries, key comparisons, borrowed figures.

MINI CASE STUDY: CODE + LICENSE

Scenario: You used a GitHub repo function in your system.

Ethical/standard steps:

- Cite the repo (and any paper associated with it)
- Check the license (MIT/Apache/GPL etc.)
- If the license requires preserving notices, do it
- Clearly state what you modified

Why it matters:

Even if it's "just a utility function," it's still intellectual work and may have legal constraints.

MINI CASE STUDY: BENCHMARK GAMING (TEST LEAKAGE)

Scenario: You tried many settings and kept checking the test accuracy.

What goes wrong:

Your test set becomes a hidden training signal → result doesn't generalize.

Fix (standard solution):

- Use validation for selection
- Lock test set until the end
- Report number of trials and selection rule
- If leakage already happened: redo with a clean split and disclose limitations

SOCIETAL IMPACT: 5 QUESTIONS FRAMEWORK

Before publishing/deploying, ask:

1. Who benefits?
2. Who could be harmed?
3. What's the misuse/dual-use scenario?
4. What hidden assumptions exist? (data coverage, context)
5. How will harms be evaluated and reduced?

note:

High accuracy can still be harmful (privacy invasion, unfair errors, unsafe behavior).

RESPONSIBLE AI / SYSTEMS RISKS

Common risks + typical mitigations:

- **Bias/Fairness:** measure performance across groups; audit dataset
- **Privacy:** minimize data; anonymize carefully; secure storage
- **Security:** threat model; defend against attacks (poisoning, prompt injection, adversarial examples)
- **Safety:** avoid harmful outputs; add guardrails; monitor misuse
- **Transparency:** document limitations; include uncertainty; explain evaluation

Mini example:

A proctoring system can falsely flag students → mitigation includes human review and careful error analysis.

SUSTAINABILITY IN CS RESEARCH

*Sustainability = achieving goals with **reasonable compute/energy cost**.*

What to report (standard, simple):

- Hardware type (e.g., GPU model)
- Training time or GPU-hours
- Number of runs / trials
- Efficiency metrics (latency, memory, throughput)

Ways to reduce compute:

- Start with small models/baselines
- Early stopping, smaller search spaces
- Reuse checkpoints
- Pruning/quantization/distillation (if relevant)

ACTIVITY: ETHICS & IMPACT STATEMENT

n pairs (10 minutes): Pick one project:

- Malware detection model
- LLM-based code autograder
- Student webcam proctoring
- Depression detection from social media

Write 6 bullets:

1. Data source + privacy risk
2. Bias/fairness risk
3. Misuse scenario
4. Mitigation plan
5. Reproducibility plan
6. Sustainability plan (compute budget + reduction strategy)

SUMMARY

Ethical CS research =

- Honest results (no fabrication/falsification/leakage)
- Fair credit (authorship + citations)
- Respect for people (privacy, consent, security)
- Transparent and reproducible methods
- Societal responsibility (bias, safety, misuse)
- Sustainable compute (report cost, reduce waste)